

Access Control Implementation

PrimeCore Minerals | Secure Multi Office Network

PC-02 | Version 0.1 | 23 September 2026

Status: Partly evidenced implementation and proposed extension

Owner: IT Administrator | Approver: Project lead | Approval date: Pending

Implement and validate role based access for PrimeCore users, shared files and network administration. This guide combines the demonstrated HQ network block with a proposed identity and file permission implementation.

Baseline and prerequisites

HQ-EMP-LAPTOP uses 192.168.11.100 on VLAN 11. HQ-MONITOR uses 192.168.90.60 on VLAN 90. R06 blocks the tested IPv4 ICMP path to management gateway 192.168.99.1, tracker 1787829015. T01 correlates four blocked requests with four central records. This does not establish application permissions or all management protocols.

Before identity changes, record the actual Samba AD domain, domain controller, file server, share paths and supported versions. Confirm recoverable identity backups, a working emergency administrator and a change window. The group names below are proposed naming conventions, not discovered directory objects.

Proposed group	Access purpose	Approver
GG_PC_Staff	Ordinary staff business access	Line manager
GG_PC_File_Project_RW	Modify approved project share	Data owner
GG_PC_File_Project_RO	Read approved project share	Data owner
GG_PC_Network_Admin	Administer network from approved management endpoint	IT owner
GG_PC_Security_Analyst	Read monitoring and incident evidence	Security owner
GG_PC_Backup_Operator	Operate backup jobs and restores	IT owner

Implementation sequence

1. Export current membership, share permissions, filesystem ACLs and firewall configuration. Record the change reference and restore procedure.
2. Create role groups and dedicated test accounts. Use named administrator accounts separate from daily accounts. Avoid direct user grants unless an approved exception requires them.
3. Map approved staff to business groups, then grant resource permissions through groups. On Samba shares, validate the interaction of share restrictions and filesystem ACLs, including inheritance and effective rights.
4. Apply network rules at the actual ingress boundary. Preserve required DNS, authentication and time services. Place the management block ahead of broad employee allow rules after reviewing floating rules and existing states.

Validation and rollback

1. Test with a fresh login or refreshed session so cached access does not conceal membership changes. Record the user, groups, target path, source address and time.
2. For each role, exercise allowed operations and one prohibited operation. A successful login alone does not prove least privilege.
3. Remove test data and temporary grants, preserve evidence, and obtain the resource owner's acceptance. Reapply exported permissions or configuration if expected access fails; verify recovery using the same tests.

Test	Expected outcome	Evidence status
AC-01 employee ICMP to 192.168.99.1	Blocked and correlated with R06	T01 passed on 13 Sep
AC-02 employee HTTPS/SSH to approved management test host	Denied and logged	To execute
AC-03 approved administrator to approved management service	Allowed from approved endpoint only	To execute
AC-04 read only account reads then writes test share	Read succeeds; write denied	To execute
AC-05 modify account creates, changes and deletes test file	Allowed only within approved share	To execute
AC-06 remove role and terminate relevant session	Further protected access denied	To execute

Test record: change ID; tester; timestamp and timezone; account and effective groups; source and destination; expected and actual outcomes; raw log and screenshot references; reviewer. Use disposable accounts and files. Do not test denial by locking the operational administrator out.

Operational handover

Record actual group names, owners, authorised administrator endpoints and exceptions in PrimeCore Access Control Policy. Reconcile membership monthly and remove obsolete access when a person changes role or leaves. Follow PrimeCore Shared File Procedure for shared-file misuse and Authentication Lockout Procedure for suspicious authentication. Retest after changes to groups, ACL inheritance, routing or firewall rule order.

Screenshot evidence appendix

These figures record selected laboratory observations. They support the tested functions described in this artefact. Proposed policies, untested controls and organisational approval remain separate from the pictured results.

Figure E1 Employee WiFi firewall rules

The screenshot shows the pfSense web interface for configuring firewall rules. The breadcrumb trail is "Firewall / Rules / EMPLOYEE_WIFI_VLAN". The "EMPLOYEE_WIFI_VLAN" tab is selected in the top navigation bar. Below the tabs, a table lists the firewall rules. The rules are ordered as follows:

States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☐	0/0 B	IPv4 *	EMPLOYEE_WIFI_VLAN subnets	*	MANAGEMENT_NETWORK	*	*	none	R06 - Block Employee WiFi Access to Management	
☐	0/0 B	IPv4 ICMP	EMPLOYEE_WIFI_VLAN subnets	*	*	*	*	none	TEMP - Employee wifi connection	
☐	0/16.00 MIB	IPv4 TCP	EMPLOYEE_WIFI_VLAN subnets	*	*	WEB_PORTS	*	none	R03a - Allow Employee WiFi Web Access	
☐	0/0 B	IPv4 TCP	EMPLOYEE_WIFI_VLAN subnets	*	PRINTER_NETWORK	PRINT_PORTS	*	none	R09b - Allow Employee WiFi for Printers	
☐	0/895 KIB	IPv4 TCP/UDP	EMPLOYEE_WIFI_VLAN subnets	*	*	DNS_PORT	*	none	R03b - Allow Employee WiFi DNS	
☐	1/2.34 MIB	IPv4 TCP	EMPLOYEE_WIFI	*	AD_SERVER	AD_TCP_PORTS	*	none		
☐	0/0 B	IPv4 UDP	EMPLOYEE_WIFI	*	AD_SERVER	AD_UDP_PORTS	*	none		

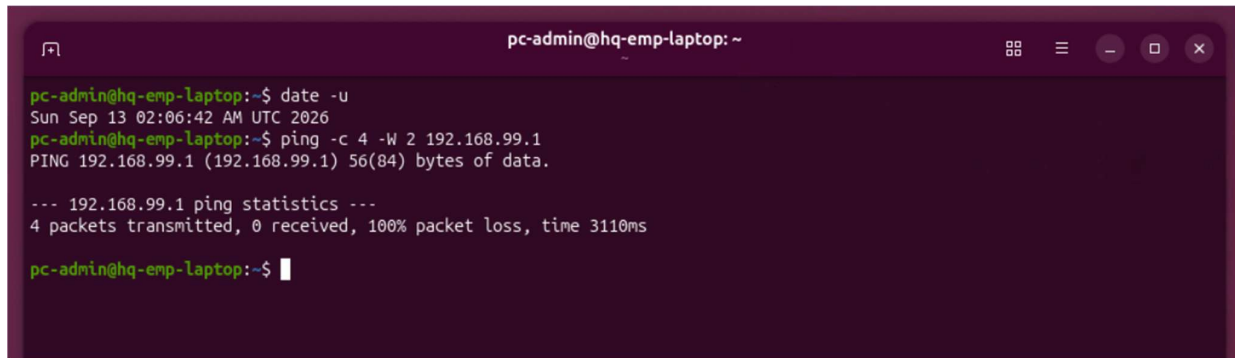
At the bottom of the table, there are buttons for "Add", "Add", "Delete", "Toggle", "Copy", "Save", and "Separator".

The displayed rule order includes R06 blocking management access. The controlled validation below covers IPv4 ICMP to 192.168.99.1, rather than every management service.

Source evidence: evidence/HQ_EMPLOYEE_WIFI_Rules.png

Screenshot evidence continued

Figure E2 Controlled management access test

A terminal window titled 'pc-admin@hq-emp-laptop: ~' with standard window controls. The terminal shows the following commands and output:

```
pc-admin@hq-emp-laptop:~$ date -u
Sun Sep 13 02:06:42 AM UTC 2026
pc-admin@hq-emp-laptop:~$ ping -c 4 -W 2 192.168.99.1
PING 192.168.99.1 (192.168.99.1) 56(84) bytes of data.

--- 192.168.99.1 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3110ms

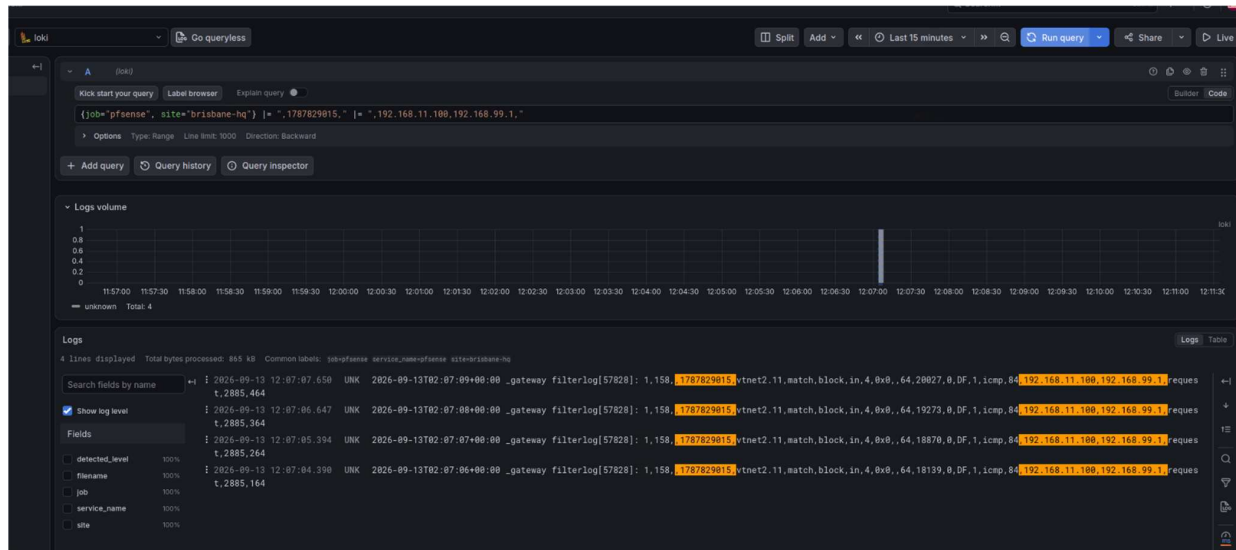
pc-admin@hq-emp-laptop:~$
```

Four ICMP requests to 192.168.99.1 received no replies. The corresponding firewall records provide the additional evidence linking this result to the tested rule.

Source evidence: evidence/T01_Employee_Ping_Blocked.png

Screenshot evidence continued

Figure E3 Correlated firewall records



Grafana retrieves four matching R06 records for source 192.168.11.100 and destination 192.168.99.1. This establishes the demonstrated collection and retrieval path.

Source evidence: evidence/T01_Grafana_R06_Four_Blocks.png